Introduction to Deep Learning

From Statistical Learning to Neural Networks

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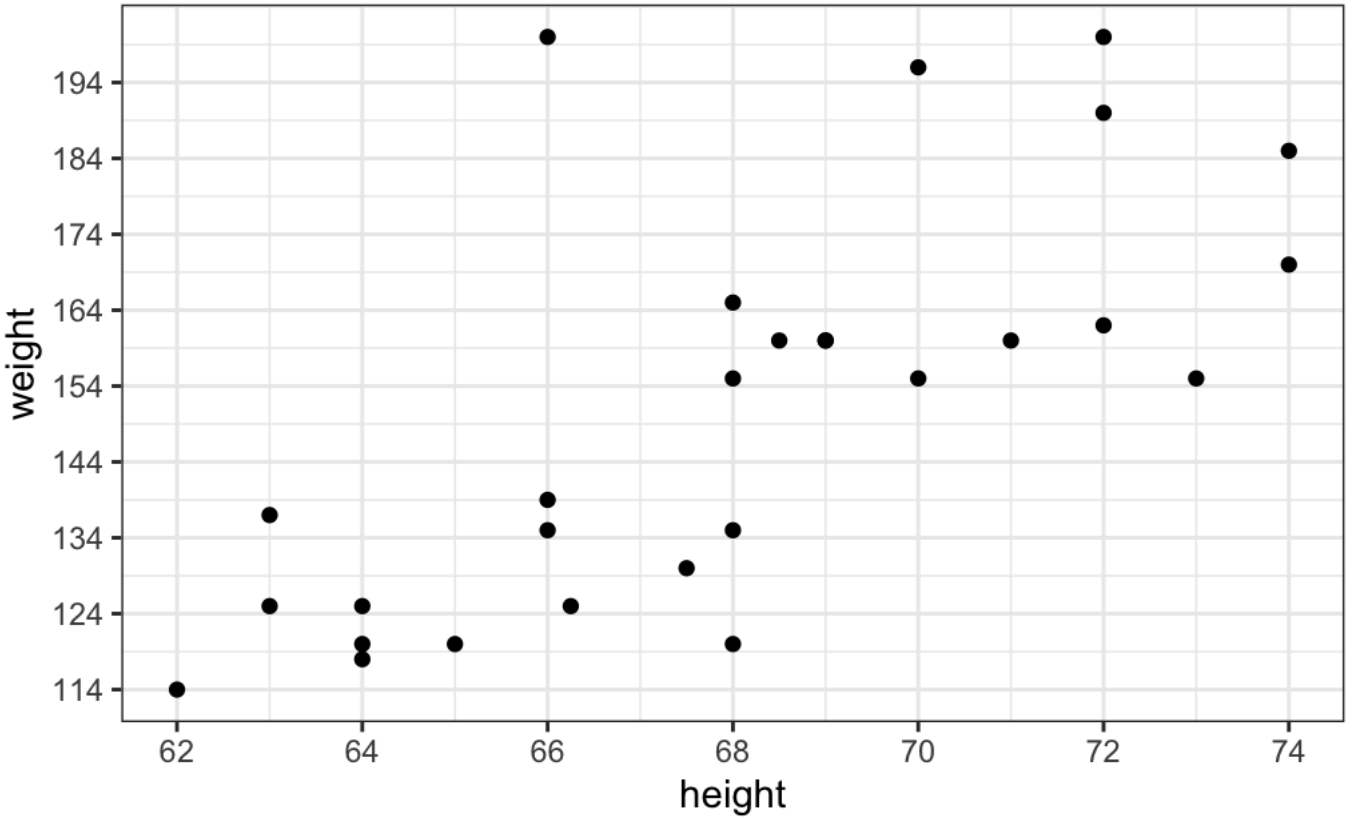
Basics of Statistical Learning

- Statistical learning serves as a basic introduction to machine learning
- Statistical learning, especially regression, can be thought of as a building block of artificial neural networks (many perceptrons)
- Learning Objectives

    1. Fit a Simple Linear Regression (SLR) model to data and interpret model coefficients (parameters)
    2. Build and train a Perceptron to solve a regression problem
    3. Build and train a deep neural network using Keras

Simple Linear Regression: Model Definition

- The goal for SLR is to investigate the relationship between the ***response*** (***Y***) and the ***predictor*** (***X***) variables
- Recall from high school algebra: $y = b + mx$
  - Where ***m*** is the slope and ***b*** is the y-intercept



Simple Linear Regression: Model Definition

- The general form of the SLR model closely resembles the algebraic equation ($\boldsymbol{y = b + mx}$):

  $$Y = \beta_0 + \beta_1 X + \epsilon$$

- For an individual observation (x_i, y_i), the regression equation becomes:

  $$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

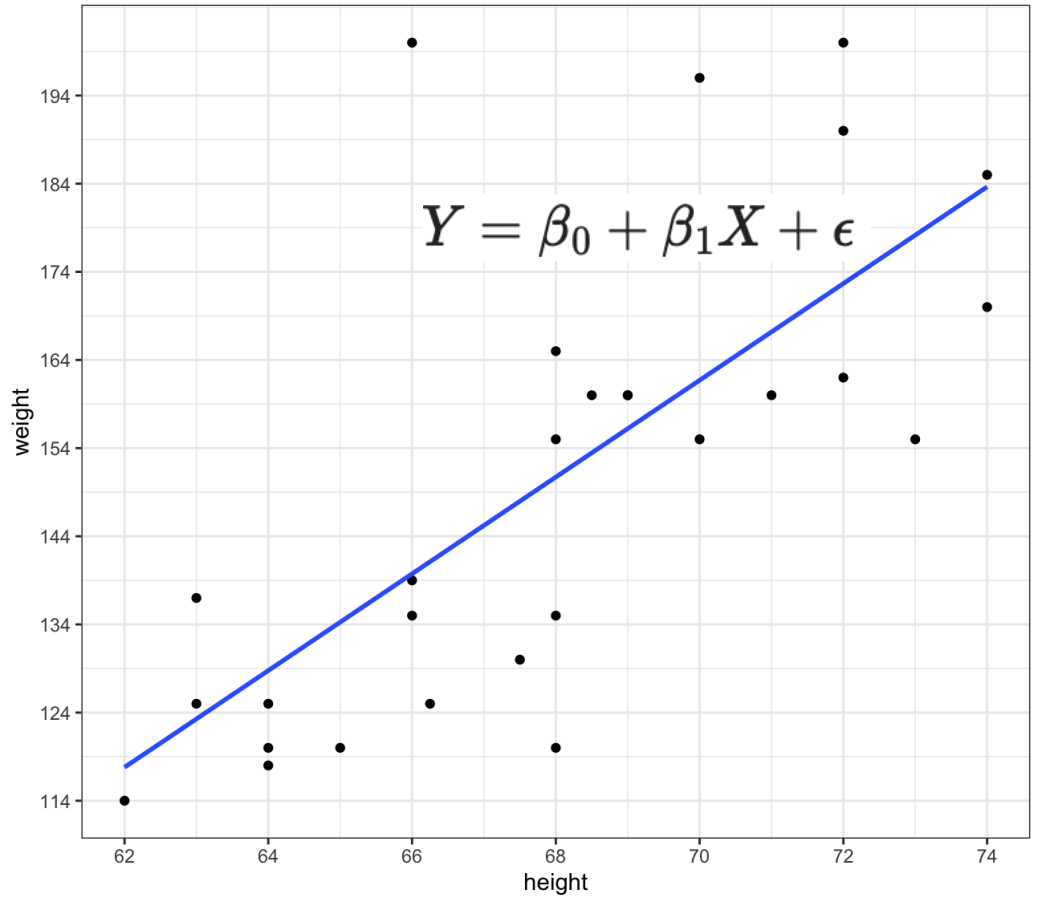
- Where
  - β_0 is the is the population y-intercept,
  - β_1 is the population slope
  - ϵ_i is the error or deviation of an observation from regression line

- Together, β_0 and β_1 are the (unknown) population model ***coefficients*** or ***parameters***

- The goal of regression is to estimate parameters ($\hat{\beta}$) to describe the relationship between ***Y*** and ***X***

- We estimate $\hat{\beta}$ using the method of Ordinary Least Squares (OLS)

Simple Linear Regression: Errors (Loss)



- Predicted value ($\hat{y}_i$) based on the $\hat{x}_i$ is obtained by

$$fit_i = \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- An error is defined by

$$res_i = \epsilon_i = y_i - \hat{y}_i$$

- We quantify the total error (loss)

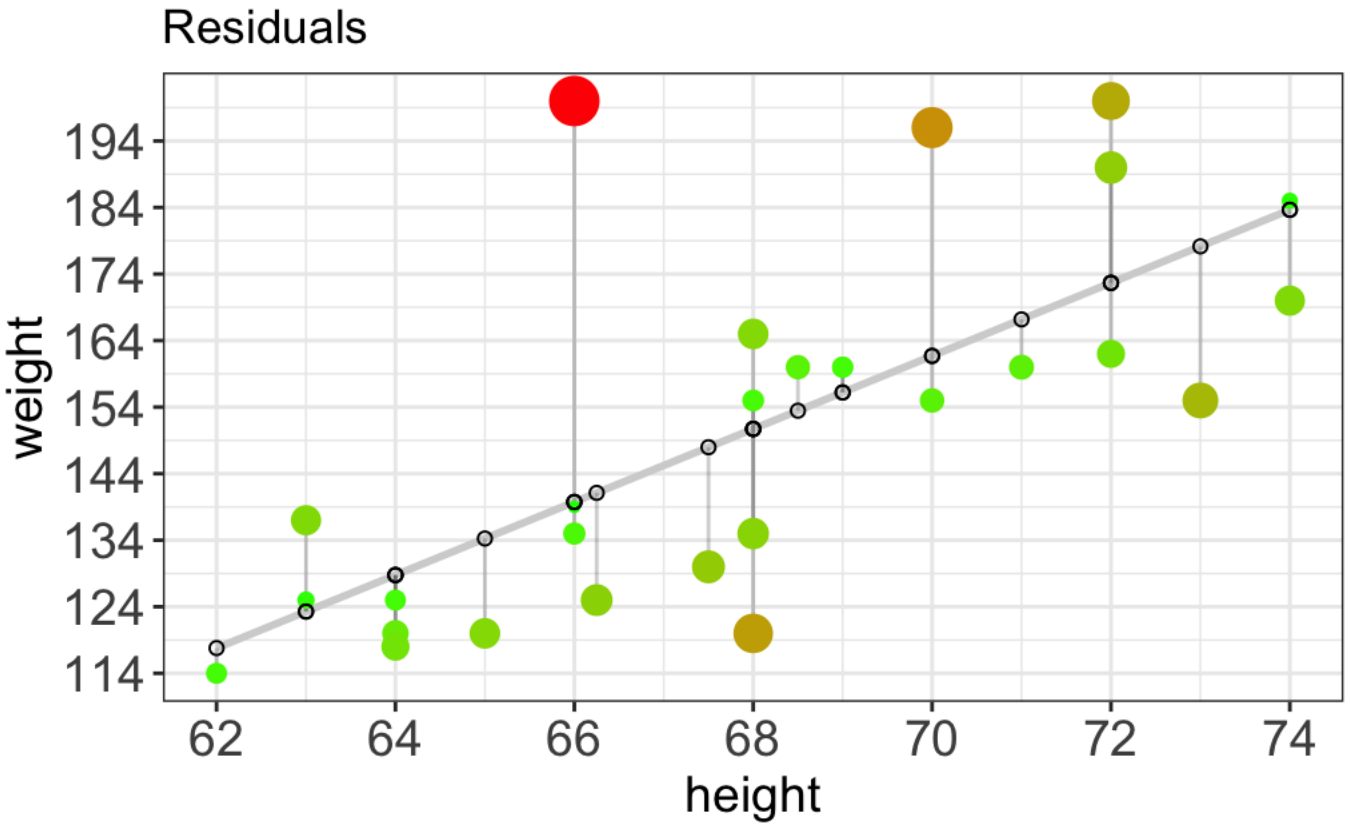
$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} \left(\hat{y}^{(i)} - y^{(i)} \right)^2$$

- Goal: obtain least squares estimates of β to minimize MSE

Simple Linear Regression: Errors (Loss)

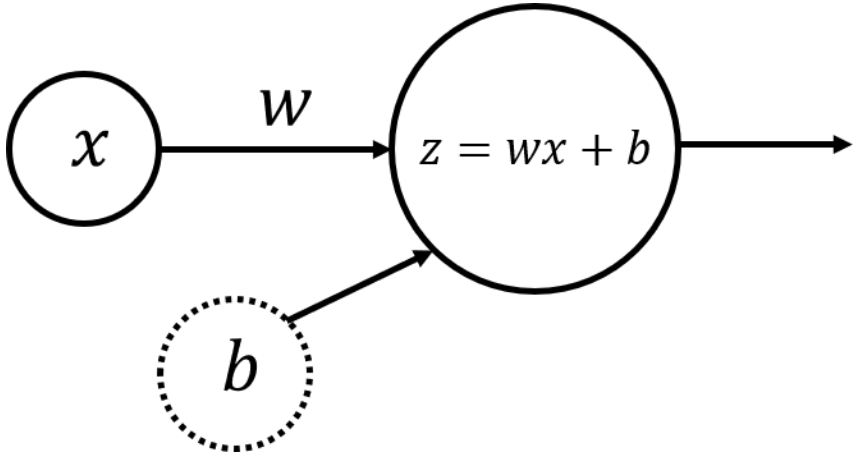
Errors, visualized



Hands-On!

Regression with Perceptron

- We will construct a ‘neuron’ corresponding to a SLR model, and train with **gradient descent**

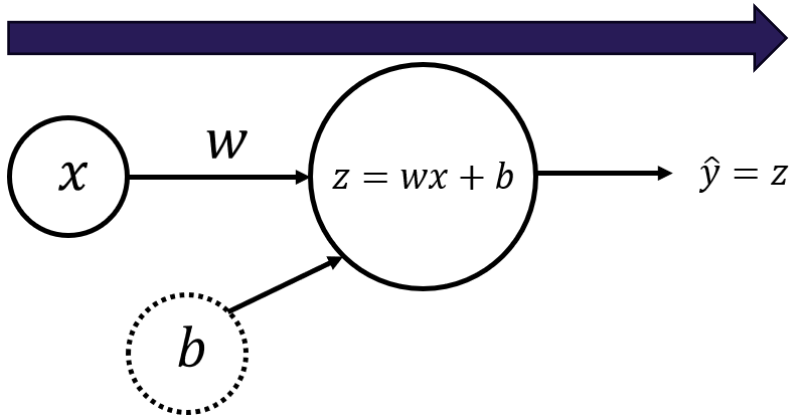


$$\hat{y} = wx + b$$

- **Weight** (*w*) and **bias** (*b*) are the parameters that will get updated when you **train** the model
- As in the previous SLR model, the goal is to identify parameters (*w & b)* that minimize the average loss function: **cost function**

$$\mathcal{L}(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(\hat{y}_i - y_i\right)^2$$

Training the model



- After defining the model structure,
  - initialize parameters to some random values (or set to 0) and
  - **update** as the training progresses
- For each training example (x_i, yi), predict $\hat{y}_i$:

$$z_i = wx_i + b,$$

$$\hat{y}_i = z_i,$$

where $i = 1, \ldots, m$

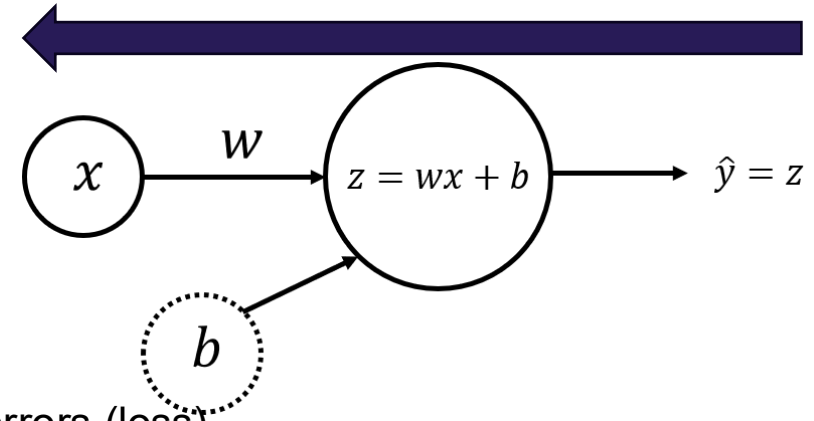
- Training examples are a vector $\boldsymbol{X}$ of size $(1 \times m)$
- Perform scalar multiplication of $\boldsymbol{X}$ by a scalar $\boldsymbol{w}$, adding $\boldsymbol{b}$.

$$Z = wX + b,$$

$$\hat{Y} = Z,$$

- This set of calculations is called **forward propagation**

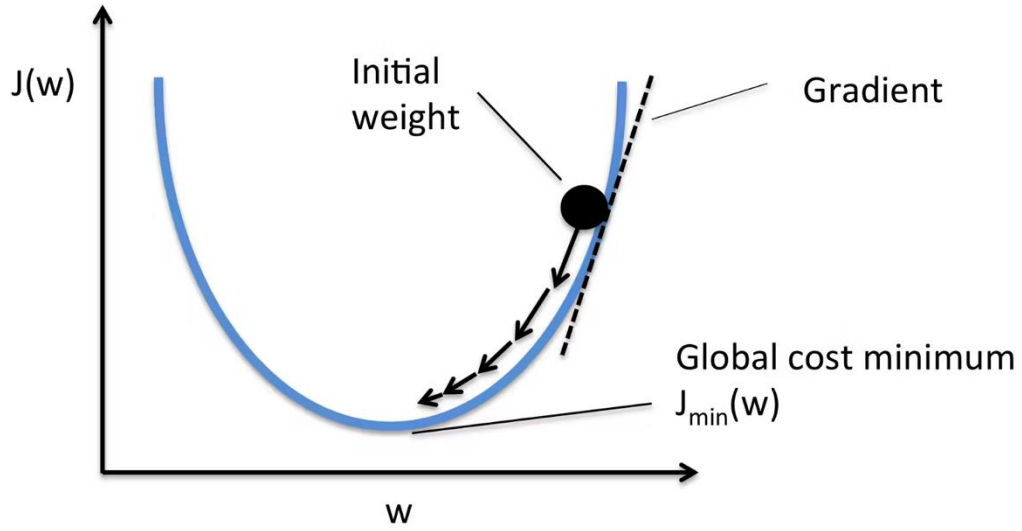
Training the model



- Given $\hat{Y}$, calculate the cost function

$$\mathcal{L}(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(\hat{y}_i - y_i\right)^2$$

  - Aim is to optimize the cost function during training, hence minimize the errors (loss)
- We minimize the cost function by use of an optimization algorithm, like **gradient descent**
  - Attempt to identify parameter values that minimize the cost function by taking partial derivatives of the cost function



We calculate partial derivatives as:

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{1}{m}\sum_{i=1}^{m}\left(\hat{y}_i - y_i\right)x_i,$$

$$\frac{\partial \mathcal{L}}{\partial b} = \frac{1}{m}\sum_{i=1}^{m}\left(\hat{y}_i - y_i\right)$$

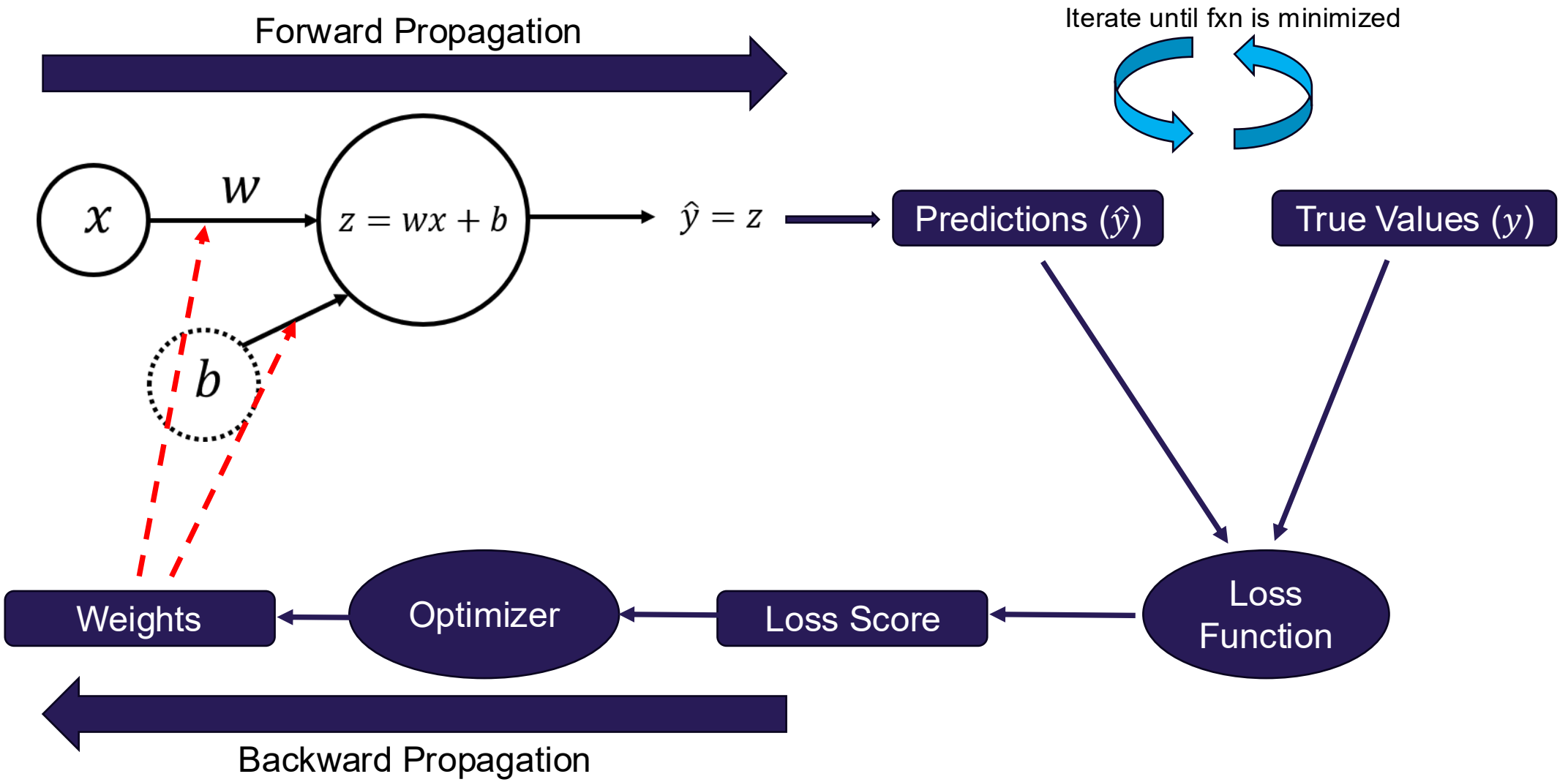
We then update the parameters iteratively using the expressions:

$$w = w - \alpha\frac{\partial \mathcal{L}}{\partial w},$$

$$b = b - \alpha\frac{\partial \mathcal{L}}{\partial b}, \quad \text{where } \alpha \text{ is the learning rate}$$

- This set of calculations is called **backward propagation**

Training the model



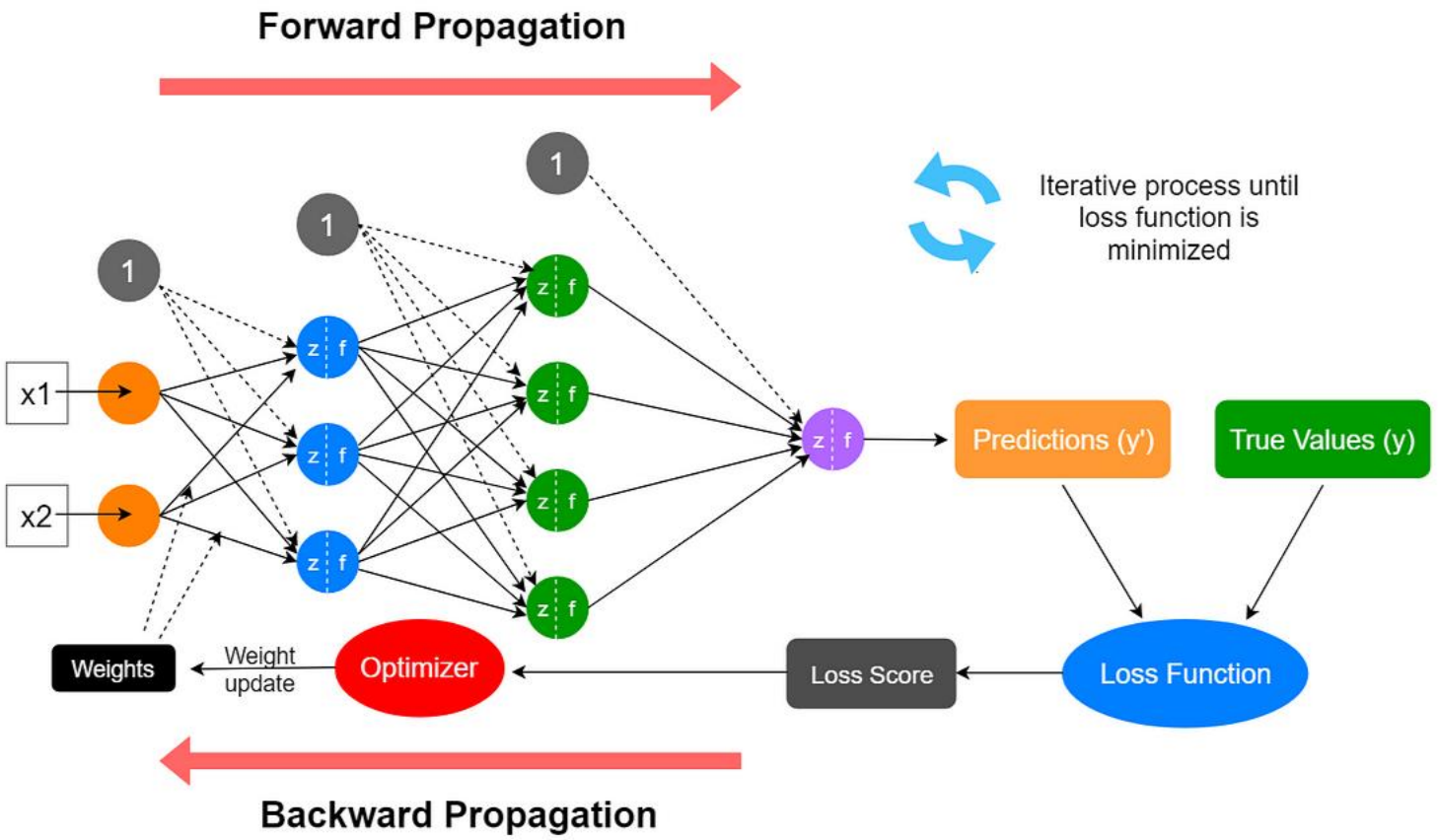
Methodology to build a neural network

- Define the neural network structure (# of input units, # of hidden units, etc)
- Initialize the model's parameters
- Loop (iterate)
  - Implement forward propagation (calculate the perceptron output)
  - Implement backward propagation (to get the required corrections for the parameters)
  - Update parameters
- Make predictions with “best” parameters

Hands-On!

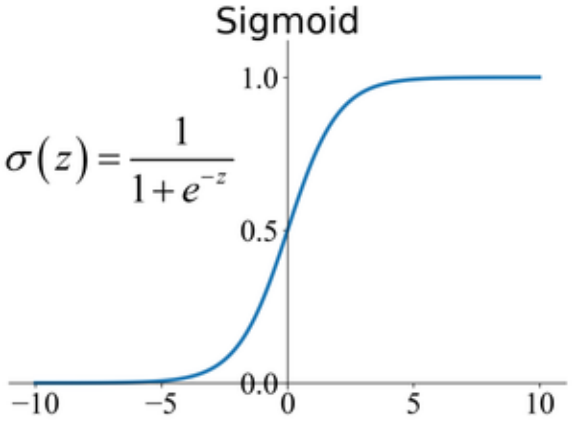
Let's Dive Deep! An Introduction to Deep Learning

The ***deep*** in deep learning: successive layers of data representation

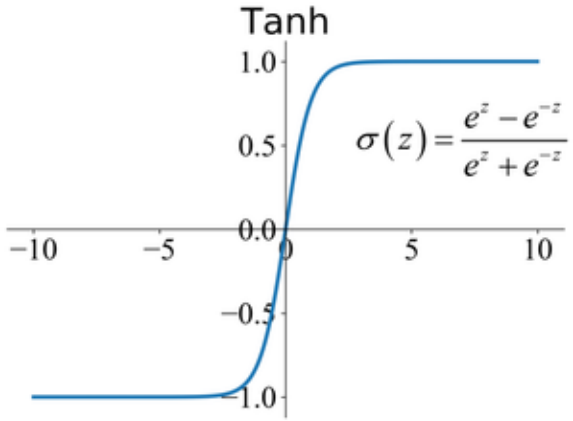


Let's Dive Deep! An Introduction to Deep Learning

Commonly used activation functions



(a)



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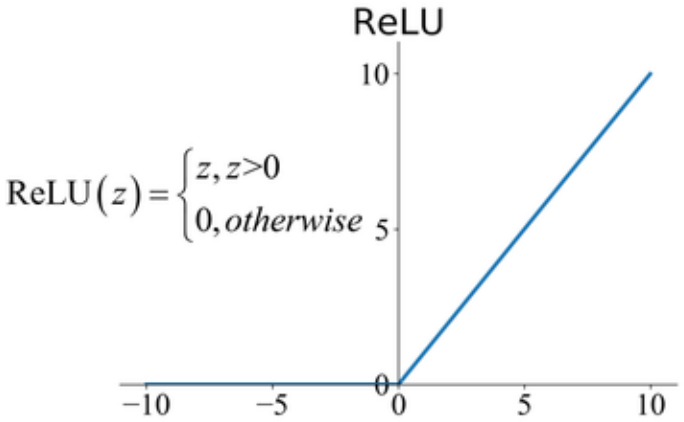

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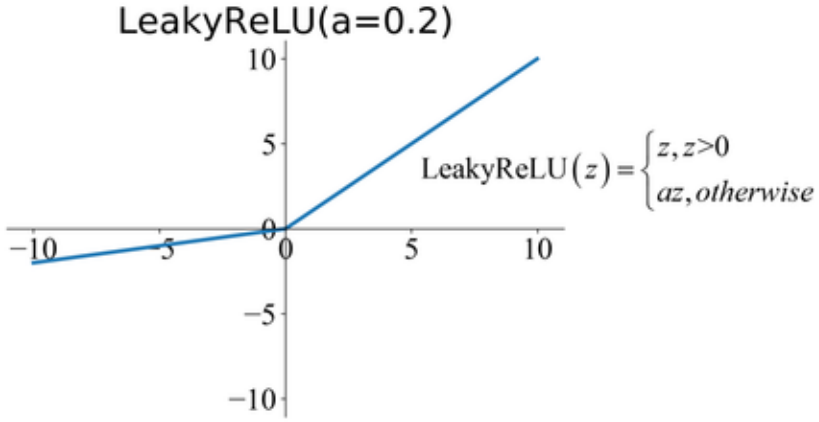

(d)

Hands-On!

Next Steps in AI: CNNs, RNNs, and LLMs

As we move beyond simple dense networks, modern AI leverages specialized architectures to tackle complex data types and tasks

Convolutional Neural Networks (CNNs)

- **Purpose:** Designed for **image and spatial data**.

- **Key Idea:** Instead of fully connecting every pixel, CNNs use **convolutional filters** to detect local patterns (edges, textures) and build hierarchical feature maps.

- **Applications:**
  - Image classification (e.g., recognizing objects in photos)
  - Medical imaging
  - Computer vision tasks like segmentation and detection

- **Why important:** CNNs exploit spatial structure, making them far more efficient and accurate for vision tasks than dense networks.

Recurrent Neural Networks (RNNs)

- **Purpose:** Handle **sequential data** where order matters.
- **Key Idea:** RNNs maintain a **hidden state** that carries information across time steps, enabling modeling of temporal dependencies.
- **Variants:**
  - **LSTM (Long Short-Term Memory)** and **GRU (Gated Recurrent Unit)** solve the vanishing gradient problem.
- **Applications:**
  - Natural language processing (NLP)
  - Time-series forecasting
  - Speech recognition
- **Why important:** RNNs introduced the ability to learn from sequences, paving the way for language models.

Large Language Models (LLMs)

- **Purpose:** Power modern AI systems for **text understanding and generation**
- **Key Idea:** Built on **Transformers**, which use **attention mechanisms** to capture relationships between words regardless of position
- **Examples:** GPT, BERT, PaLM
- **Applications:**
  - Chatbots and virtual assistants
  - Code generation
  - Summarization and translation
- **Why important:** LLMs represent a leap in scale and capability, enabling AI systems to perform reasoning, dialogue, and creative tasks.

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Questions?